

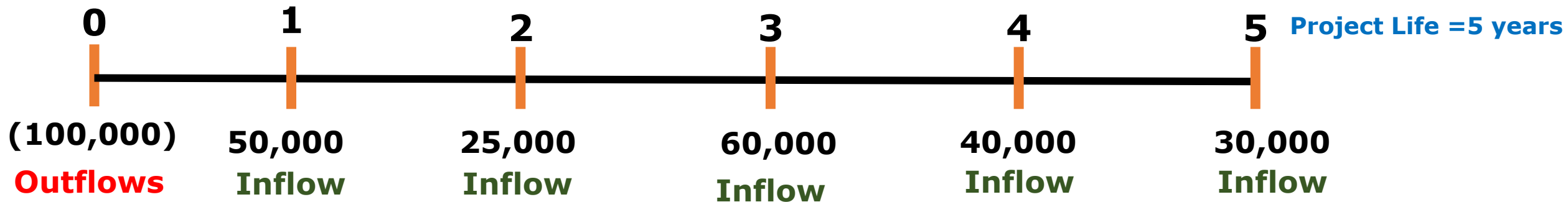
Capital Budgeting



Project Financing



Capital Budgeting is a long term investment decision where present value of all the inflows generated over project life is compared with the outflow of project so as to determine its viability.



Cash Flow Chart:

Cash Flow Chart is a chart showing the Inflows and outflows of cash for a Business in Financial Year.

Capital Budgeting decision is taken on the following parameters:

- ✓ Net Present Value (NPV)
- ✓ Profitability Index (PI)
- ✓ Payback Period (PB)
- ✓ Discounted Payback Period
- ✓ Internal Rate of Return (IRR)
- Return on Capital Employed

Basic Concept

Present Value and Future Value

Example of Future Value: How much money will I have one year from now if I invest \$ 100 at an expected 20% annul return?

$$\text{\$ 100} \xrightarrow{\times 1.2} \text{\$ 120}$$

\$ 100 Multiplied by 1.2 is \$ 120

So to get to the future value, you move from left to right: take a present value multiply it by 1 plus rate of return

Example of Present Value: How much money do I invest today to achieve \$ 120 one year from now at an expected 20% annul return?

$$\text{\$ 100} \xleftarrow{/1.2} \text{\$ 120}$$

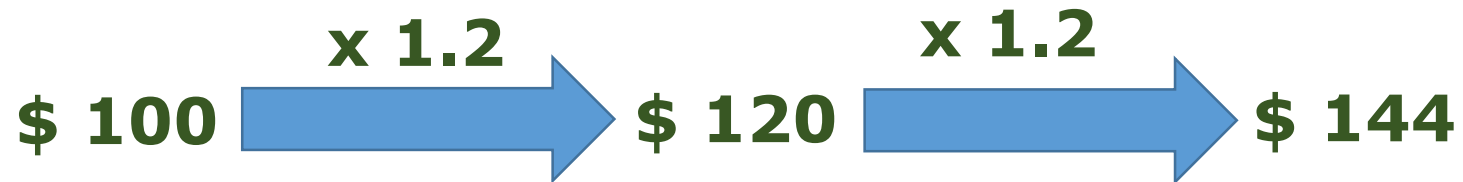
\$ 120 divided by 1.2 is \$ 100

To get to the Present Value, you go from right to left: take a future value, and divided it by 1 plus rate of turned

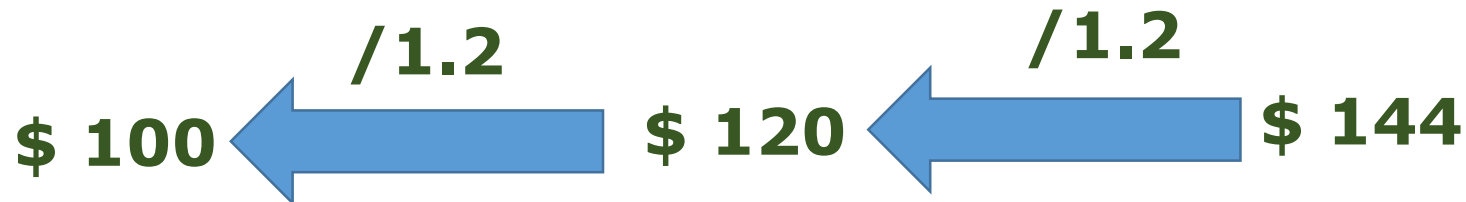
Basic Concept

Present Value and Future Value

Example of Future Value: How much money will I have two years from now if I invest \$ 100 at an expected 20% annul return?

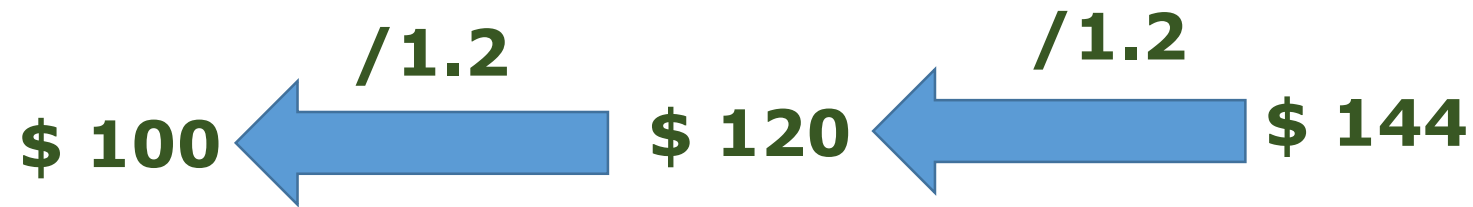
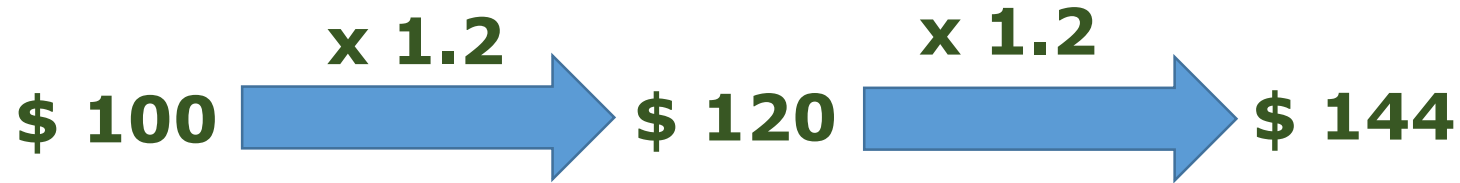


Example of Present Value: How much money do I invest today to achieve \$ 144 two years from now at an expected 20% annul return?



Basic Concept

Present Value and Future Value



Formula of Future Value

$$FV = PV (1 + r)^n$$

$$FV = 100 (1 + 20\%)^1$$

$$FV = 100 (1 + 0.2)^1$$

$$FV = 100 (1.2)^1$$

$$FV = 120$$

$$FV = PV (1 + r)^n$$

$$FV = 100 (1 + 20\%)^2$$

$$FV = 100 (1 + 0.2)^2$$

$$FV = 100 (1.2)^2$$

$$FV = 144$$

Formula of Present Value

$$PV = \frac{FV}{(1 + r)^n}$$

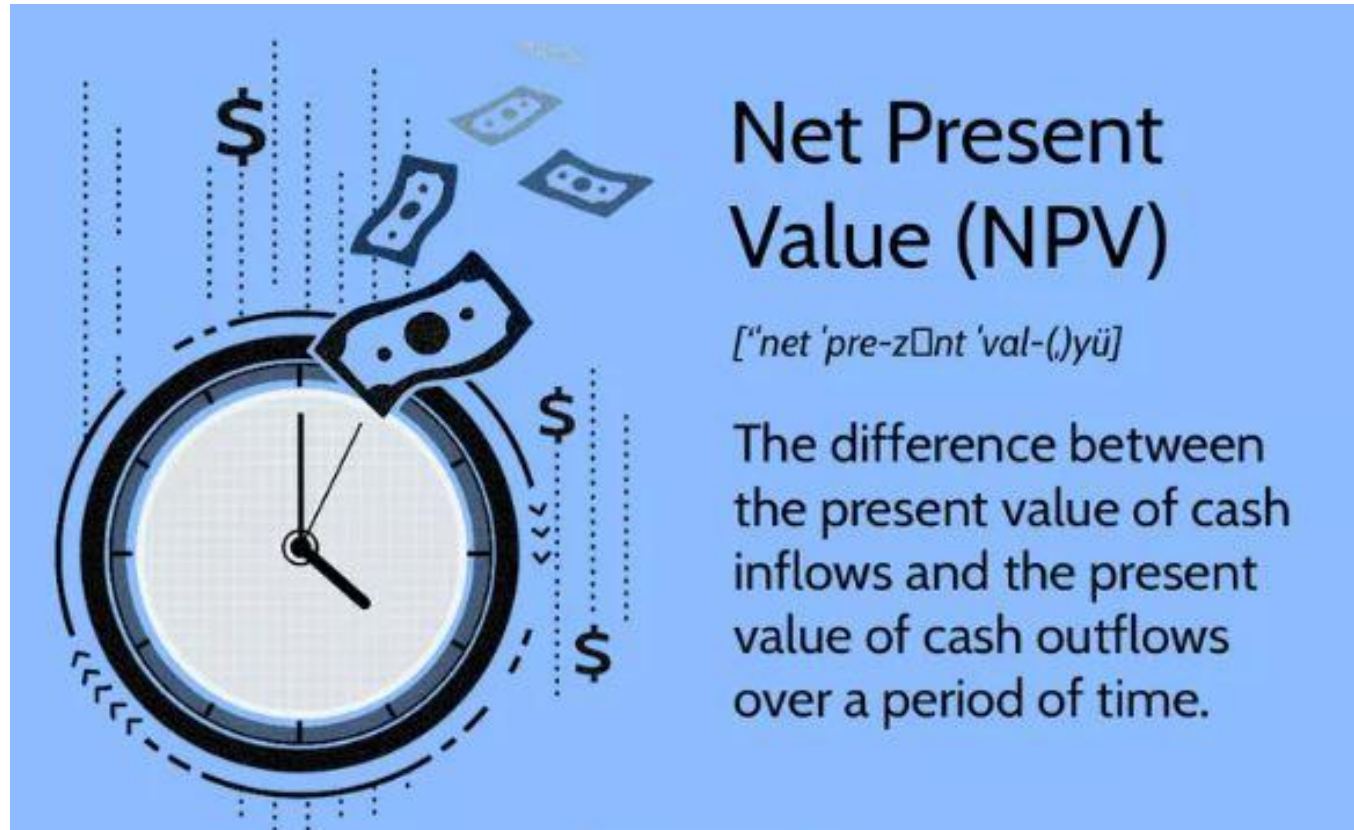
$$PV = \frac{144}{(1 + 20\%)^2}$$

$$PV = \frac{144}{(1 + 0.22)^2}$$

$$PV = \frac{144}{(1.20)^2}$$

$$PV = 100$$

Net Present Value (NPV)



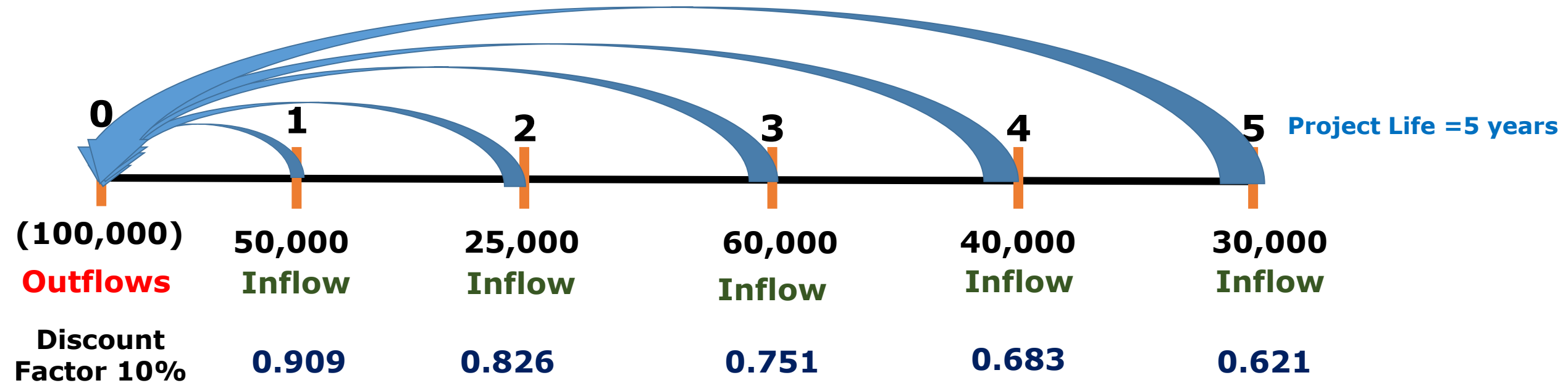
What Is Net Present Value (NPV)?

- Net present value (NPV) is the difference between the present value of cash inflows and the present [value](#) of cash outflows over a period of time.
- NPV is used in [capital budgeting](#) and investment planning to analyze the profitability of a projected investment or project.

Net Present Value (NPV)

Net Present Value (NPV) means surplus of Present Value of Inflow generated over Outflow.

In other words surplus of Inflow over and above Outflow and Cost of Outflow is NPV.

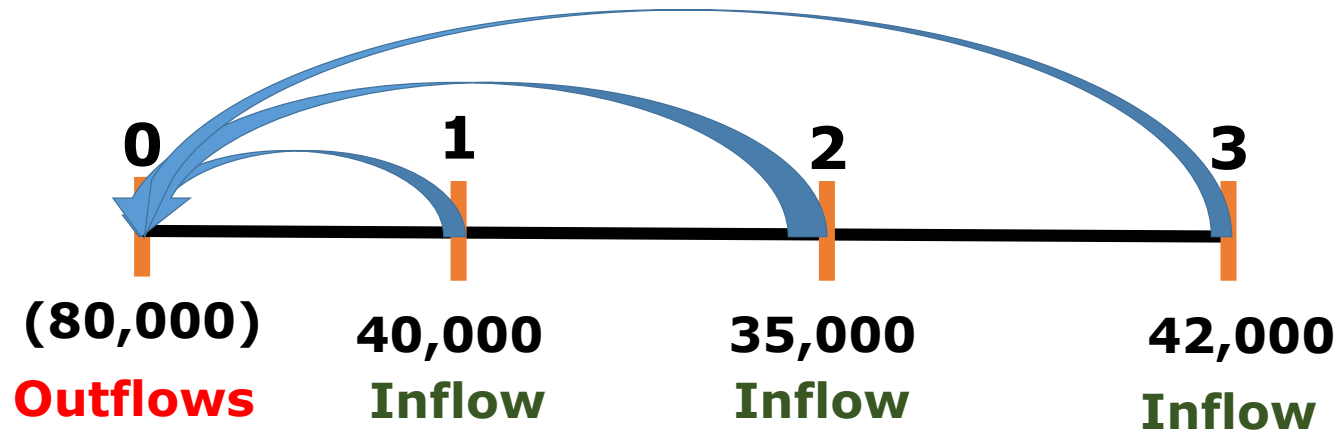


Formula of Discount Factor

$$\text{Discount factor} = \frac{1}{(1 + r)^n}$$

Net Present Value (NPV)

For Example



Formula of NPV

NPV = PV of Inflow – PV of Outflow

NPV = 96,812 – 80,000 = Rs. 16,812

Year	Inflow (1)	PV Factor @ 10% (2)	PV (1 x 2)
1	40,000	0.909	36,360
2	35,000	0.826	28,910
3	42,000	0.751	31,542
	117,000		96,812

PV Discount Factor

$$\text{Discount factor} = \frac{1}{(1 + r)^n}$$

$$\text{Discount factor} = \frac{1}{(1 + 10\%)}$$

$$\text{Discount factor} = \frac{1}{(1.1)}$$

$$\text{Discount factor} = 0.909$$

Example:

Calculate NPV
Project Life = 5 years
Outflow = 400,000

Year	Inflow
1	100,000
2	300,000
3	150,000
4	250,000
5	50,000

Cost of Outflow = 10%
Scrap Value = 5,000

Solution

PV of Inflow

Year	Inflow (1)	PV Factor @ 10% (2)	PV (1 x 2)
1	100,000	0.909	90,900
2	300,000	0.826	247,800
3	150,000	0.751	112,650
4	250,000	0.683	170,750
5	(5,000+50,000)=55,000	0.621	34,155
			656,255

Formula of NPV

NPV = PV of Inflow – PV of Outflow

NPV = 656,255 – 400,000 = **Rs. 253,150**

Decision Based on NPV

NPV	Decision
+ve	Accept the Project
-ve	Decline or Reject the Project
Nil	Accept/Decline

Example:

Calculate NPV for following given values:

Initial Investment \$ 100,000
Cash in flow /year \$ 25,000
Life 6 years
i 10 %

Solution

Year	Cash Flow (CF) \$	Present Value (PV)
0	$\frac{-100,000}{(1 + 0.10)^0}$	\$ -100,000
1	$\frac{25,000}{(1 + 0.10)^1}$	22,727
2	$\frac{25,000}{(1 + 0.10)^2}$	20,661
3	$\frac{25,000}{(1 + 0.10)^3}$	18,783
4	$\frac{25,000}{(1 + 0.10)^4}$	17,075
5	$\frac{25,000}{(1 + 0.10)^5}$	15,523
6	$\frac{25,000}{(1 + 0.10)^6}$	14,112

Formula of NPV

NPV = PV of Inflow – PV of Outflow

NPV = 108,881 – 100,000 = **Rs. 8,881**

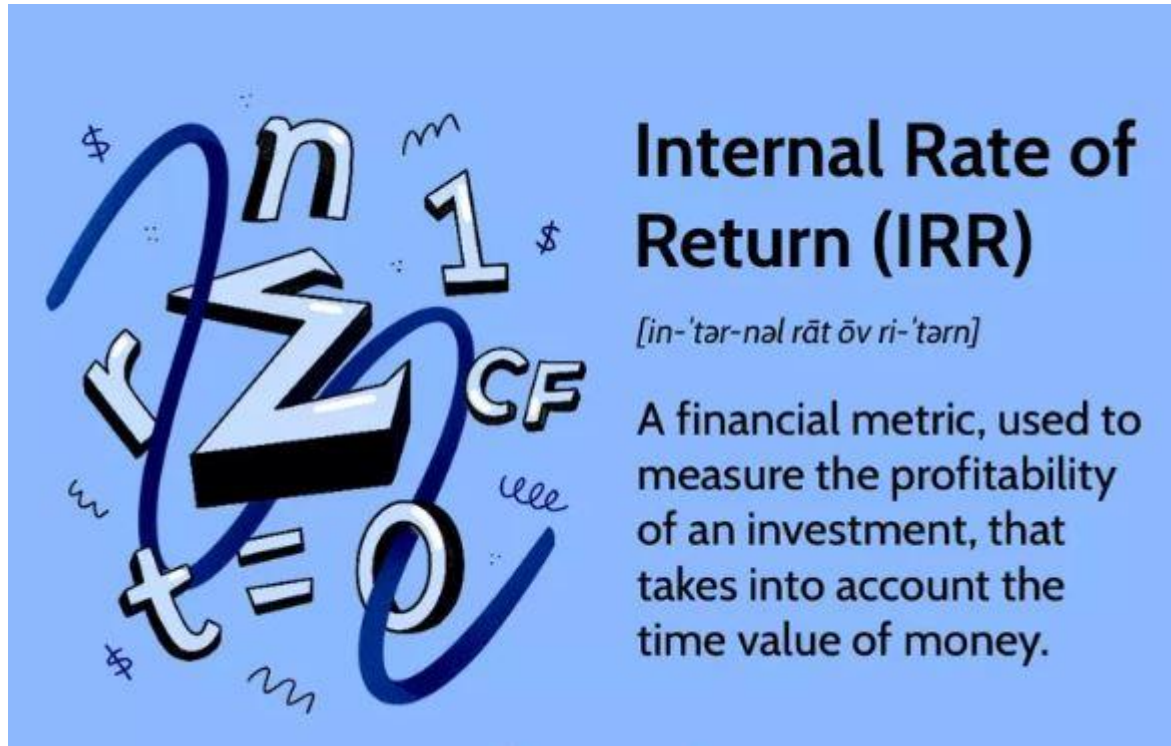
The result is +ve

Probability Index = $\frac{\text{PV of Inflows}}{\text{PV of Outflows}}$

Profitability Index (PI) = $\frac{108,881}{100.000} = 1.08$

It PI should be greater than one, for project to be desirable to invest.

Internal Rate of Return (IRR)



What Is Internal Rate of Return (IRR)?

The internal rate of return (IRR) is a metric used in financial analysis to estimate the profitability of potential investments. IRR is a [discount rate](#) that makes the net present value (NPV) of all cash flows equal to zero in a discounted cash flow analysis.

Internal Rate of Return (IRR)

- IRR or irr is the annual rate of return on investment
- IRR is expressed in % age
- The higher the IRR the better it is.
- If there are 2 or more projects their project having higher IRR is recommended.
- In IRR a hurdle rate is assumed.
- Hurdle rate is assumed by investor and it depends upon expected rate of return from project from investment, but also considers the nature of project.
- Some projects have higher yield and higher rate of return like road project in improvement is visible and user charges can be collected from beneficiaries more easily. Through toll tax. ROR is over 10%.

Internal Rate of Return (IRR)

However the r.o.r from water or sewerage project is lower since work is underground and visible improvements in water supply depends upon adequate equitable water distribution and quality of water confirming to WHO Guidelines.

ROR is less than 3%

Formula IRR

$$IRR = Low\ Rate + \frac{NPV\ at\ Low\ Rate}{NPV\ at\ Low\ Rate - NPV\ at\ High\ Rate} \times Difference\ in\ Rate$$

$$IRR = A\% + \frac{a}{a - b} \times (B\% - A\%)$$

IRR is calculated by Hit and Trail Method.

Problem 1 Project A has an initial outlay of \$ 20,000 with \$ 4000 cash in-flow per year. The project has 8 years life. Interest Rate is 6% & SV is zero.

Project B has an initial outlay of \$ 25,000 with \$ 5000 cash in-flow per year. Project B also has 8 years life. Project B. Interest Rate is 10% & SV \$ 5000.

Calculate NPV and IRR for both projects.

Assume Hurdle Rate of 15%.

- a) Determine NPV and IRR of Project A and Project B
- b) Do you recommended Project A or Project B.

Project 1.

Given:

	<u>Project A</u>	<u>Project B</u>
Initial Cost of Project	\$ 20,000	\$ 25,000
i %	6 %	10 %
Cash flow / year	\$ 4000 / year	\$ 5,000 /year
Period “n”	8 years	8 years
Salvage Value	Zero	\$ 5,000
Hurdle Rate	15%	15%

Solution: Substitute values in Equation (1), (2) & (3)

Formulae

Project A

1.
$$NPV_{i \%} = A \left[\frac{(1 + i \%)^N - 1}{i (1 + i \%)^N} \right] - I \dots\dots\dots (1)$$

i % is initial interest rate

NPV with SV

2.
$$NPV_{i \%} = A \left[\frac{(1 + i \%)^N - 1}{i (1 + i \%)^N} \right] - I + \frac{SV}{(1 + i \%)^N} \dots\dots\dots (2)$$

3.
$$IRR = A \% + \frac{a}{a - b} (B \% - A \%) \dots\dots\dots (3)$$

Where

- A % = Low income rate %
- B % = Assumed higher income rate %
- a = + ve NPV
- b = - ve NPV

Substitute values in equation (1)

$$\begin{aligned}\text{NPV}_{6\%} &= 4000 \left[\frac{(1 + 0.06)^8 - 1}{0.06 (1 + 0.06)^8} \right] - 20,000 \\ \text{NPV}_{6\%} &= \$ 4839.2\end{aligned}$$

Assume another higher income rate $i\%$ say 15 %

Substitute values in equation (2)

$$\begin{aligned}\text{NPV}_{15\%} &= 4000 \left[\frac{(1 + 0.15)^8 - 1}{0.15 (1 + 0.15)^8} \right] - 20,000 \\ \text{NPV}_{15\%} &= - 2050.7\end{aligned}$$

Substitute values in equation 3

$$1/\text{IRR} \% = 6\% + \frac{4839.2}{4839.2 + 2050.7} (15\% - 6\%)$$

$$\text{IRR} = 12.32\%$$

Project B

Substitute Values in equation 2

$$\text{NPV}_{10\%} = 5000 \left[\frac{(1 + 0.1)^8 - 1}{0.1 (1 + 0.1)^8} \right] - 25,000 + \frac{5000}{(1.1)^8}$$

$$\text{NPV}_{10\%} = \$ 4007.17$$

Assuming $i = 15\%$

$$\text{NPV}_{15\%} = 5000 \left[\frac{(1 + 0.15)^8 - 1}{0.15 (1 + 0.1)^8} \right] - 25,000 + \frac{5000}{(1.1)^8}$$

$$\text{NPV}_{15\%} = \$ -928.88$$

Substitute values in equation 3

$$\text{IRR} = 10 \% + \frac{4007.17}{4007.17 + 928.88} (15 \% - 10 \%)$$

$$\text{IRR} = 14.06 \%$$

Project A IRR % = 12.32%

Project B IRR % = 14.06 %

Project B is recommended.

Problem 2. Management of an organization received 2 alternates for improvement of water treatment works (WTW). The given values are as follows:

<u>Given:</u>	<u>Alternate A</u>	<u>Alternate B</u>
Initial Cost of Project	= \$ 20,000	\$ 25,000
Interest Rate	= 8 %	11 %
Cash inflow per year	= \$ 4200 / year	\$ 5,200 /year
Period	= 8 years	8 years
Salvage Value	= Zero	\$ 5,000
Hurdle Rate (Assumed value)	= 15%	15%

- a) Determine NPV and IRR of Alternate A and Alternate B
- b) Do you recommended Alternate A or Alternate B and why.

Solution

<u>Alternate A</u>	<u>Alternate B</u>
I = 20,000	I = 25000
i = 8%	i = 8%
n = 8 years	n = 8 years
A = \$ 4,200	A = \$ 5,200
SV = zero	SV = 5,000
HR = 15%	HR = 15%

$$NPV_{8\%} = 4200 \frac{(1 + 8\%)^8 - 1}{8\% (1 + 8\%)^8} - 20,000$$

$$NPV_{8\%} = 4200 \left[\frac{(1 + 0.08)^8 - 1}{0.08 (1 + 0.08)^8} \right] - 20,000$$

$$NPV_{8\%} = 4200 \left[\frac{0.850}{0.08 (1.85)} \right] - 20,000$$

$$NPV_{8\%} = 4200 \left[\frac{0.850}{0.148} \right] - 20,000$$

$$NPV_{8\%} = 4200 (5.743) - 20,000$$

$$NPV_{8\%} = \$ 4120$$

At NPV 15% (assumed)

$$NPV_{15\%} = 4200 \left[\frac{(1 + 15\%)^8 - 1}{15\%(1 + 15\%)^8} \right] - 20,000 = NPV_{15\%} = 4200 \left[\frac{(1 + 0.15)^8 - 1}{0.15 (1 + 0.15)^8} \right] - 20,000$$

$$NPV_{15\%} = 4200 \left[\frac{(3.059) - 1}{0.15(3.059)} \right] - 20,000$$

$$NPV_{15\%} = 4200 \left[\frac{2.059}{0.458} \right] - 20,000$$

$$NPV_{15\%} = 4200 (4.495) - 20,000$$

$$NPV_{15\%} = 18,073 - 20,000$$

$$NPV_{15\%} = - \$ 1121$$

$$\text{IRR}_A = A\% + \frac{a}{a - b} (B\% - A\%)$$

$$\text{IRR}_A = 8\% + \frac{4120}{4120 + 1121} (15\% - 18\%)$$

$$\text{IRR}_A = 8\% + \frac{4120}{5241} (7\%)$$

$$\text{IRR}_A = 8\% + 0.786 (7\%)$$

$$\text{IRR}_A = 13.5\%$$

$$\text{IRR}_A = 13.5\%$$

Alternate B

$$NPV_{15\%} = 5200 \left[\frac{(1 + 15\%)^8 - 1}{8\%(1 + 15\%)^8} \right] - 25,000 + \frac{5000}{(1 + 15\%)^8}$$

$$NPV_{15\%} = 5,200 (4.487) - 25,000 + \frac{5,000}{3.059}$$

$$NPV_{15\%} = 23,332 - 25,000 + 1,634$$

$$NPV_{15\%} = \$ - 34$$

$$IRR_B = A\% + \frac{a}{a - b} (B\% - A\%)$$

$$IRR_B = 11\% + \frac{3530}{3530 + 34} (15\% - 11\%)$$

$$IRR_B = 11\% + 0.99 (4)$$

$$IRR_B = 11\% + 3.96\%$$

$$IRR_B = 14.96\%$$

$$IRR_B = 15\%$$

Profitability Index

The ratio between the present value of future cash flows to the initial investment

What is the Profitability Index?

The Profitability Index (PI) measures the ratio between the present value of future cash flows and the initial investment. The index is a useful tool for ranking investment projects and showing the value created per unit of investment.

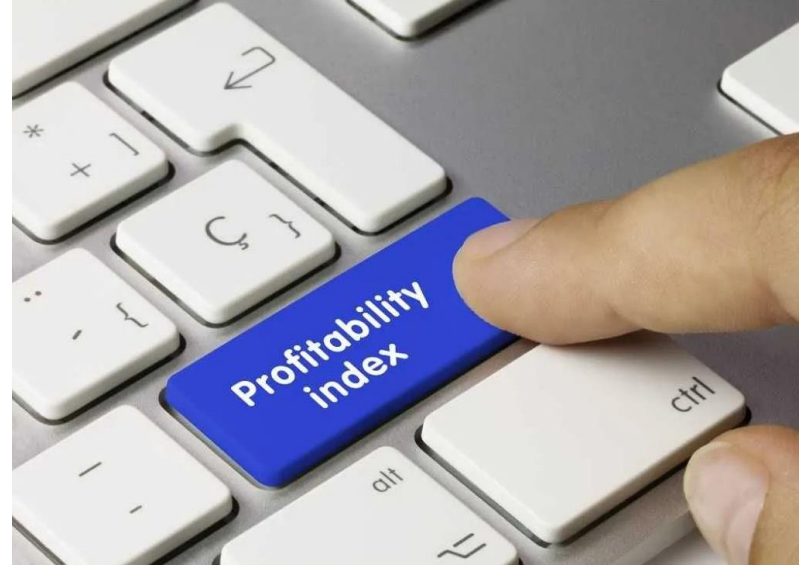
The Profitability Index is also known as the Profit Investment Ratio (PIR) or the Value Investment Ratio (VIR).

Profitability Index Formula

$$\text{Profitability Index} = \frac{\text{Present Value of Future Cash Flows}}{\text{Initial investment}}$$

OR

$$\text{Probability Index} = \frac{\text{PV of Inflows}}{\text{PV of Outflows}}$$



Therefore:

- If the PI is greater than 1, the project generates value and the company may want to proceed with the project.
- If the PI is less than 1, the project destroys value and the company should not proceed with the project.
- If the PI is equal to 1, the project breaks even and the company is indifferent between proceeding or not proceeding with the project.

The higher the profitability index, the more attractive the investment.

For Example: Two Projects A and B

	Project A	Project B
Outflows	Rs. 100,000	Rs. 80,000
Inflows	Rs. 120,000	Rs. 100,000
NPV	Rs. 20,000	Rs. 20,000

Calculate Profitability Index (PI)

Project A

$$\text{Probability Index} = \frac{\text{PV of Inflows}}{\text{PV of Outflows}}$$

$$\text{Probability Index} = \frac{120,000}{100,000}$$

$$\text{Probability Index} = 1.20$$

Project B

$$\text{Probability Index} = \frac{\text{PV of Inflows}}{\text{PV of Outflows}}$$

$$\text{Probability Index} = \frac{100,000}{80,000}$$

$$\text{Probability Index} = 1.25 \checkmark$$

Project B is better and Accept B over A

Return on Capital Employed

Returned earned on Project after considering all expenses including depreciation and taxes.

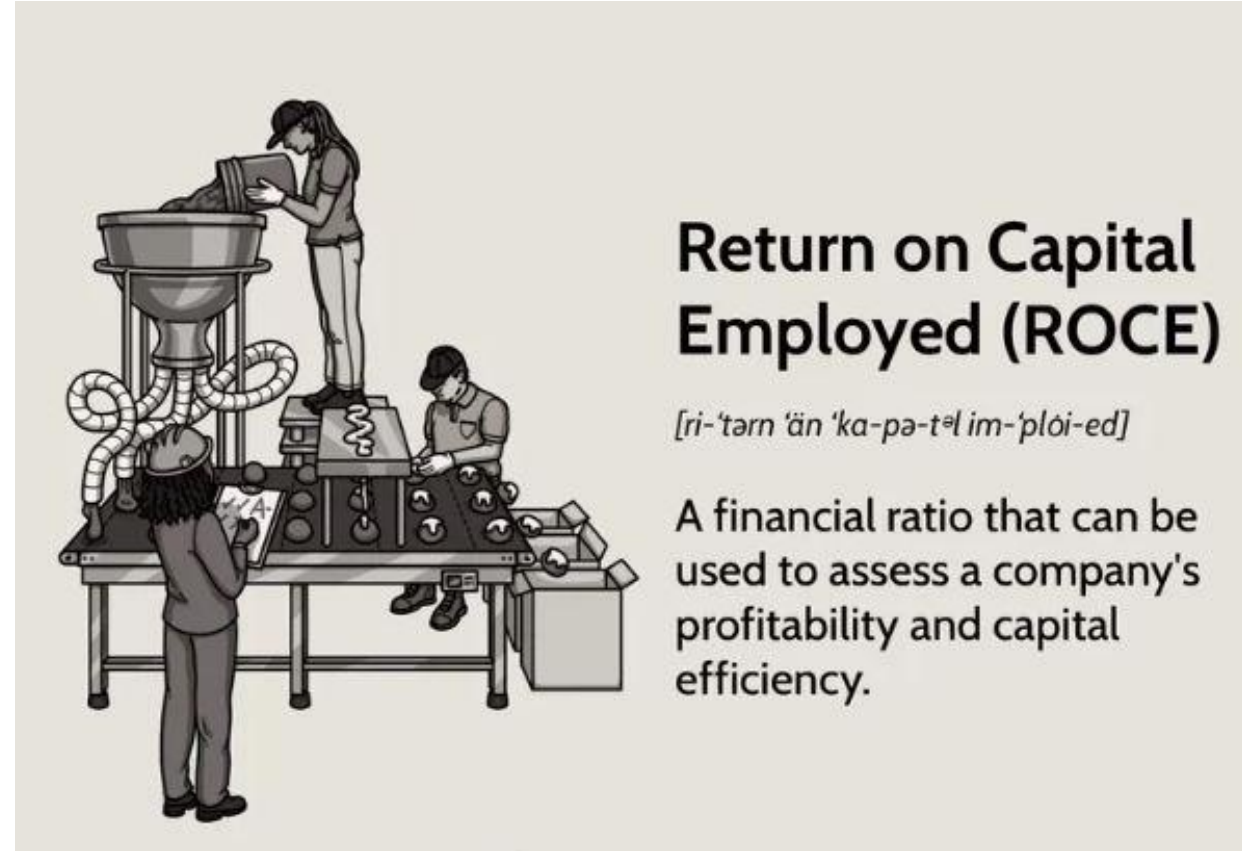
1. On Initial Capital Employed

$$\text{Returned on CE} = \frac{\text{Average PAT}}{\text{Initial Capital Employed}}$$

2. On Average Capital Employed

$$\text{Returned on CE} = \frac{\text{Average PAT}}{\text{Average Capital Employed}}$$

$$\text{Average Capital Employed} = \frac{\text{Initial Capital Employed} - \text{Scrap Value}}{2}$$

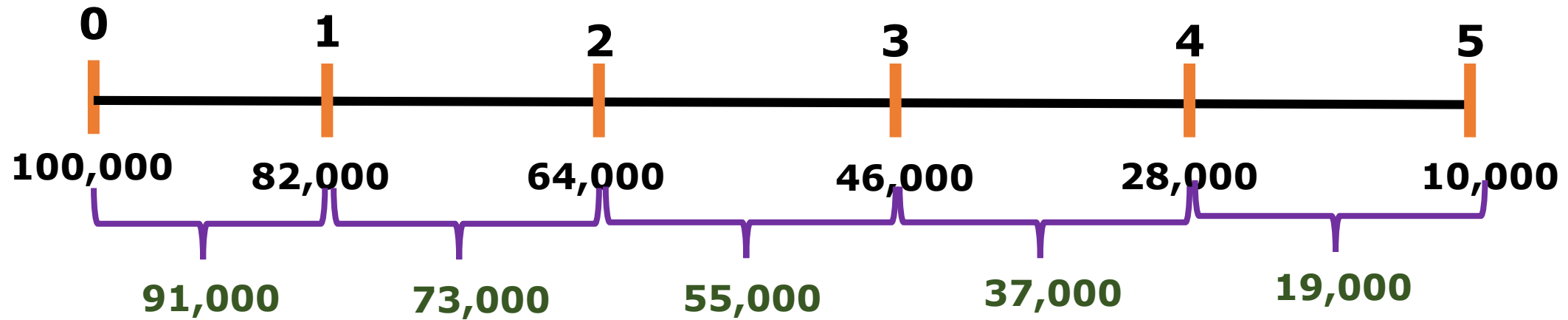


Return on Capital Employed (ROCE)

[ri-'tərn 'än 'ka-pə-təl im-'plɔi-ed]

A financial ratio that can be used to assess a company's profitability and capital efficiency.

Example. 1. Suppose that a Company has outflow, inflows and project life are given below:



$$\frac{100,000 + 82,000}{2} = 91,000$$

$$\text{Average Capital Employed} = \frac{91,000 + 73,000 + 55,000 + 37,000 + 19,000}{5}$$

$$\text{Average Capital Employed} = 55,000$$

$$\text{Average Capital Employed} = \frac{\text{Initial Capital Employed} - \text{Scrap Value}}{2}$$

$$\text{Average Capital Employed} = \frac{100,000 - 10,000}{2}$$

$$\text{Average Capital Employed} = 55,000$$

Return on Investment (ROI)

Return on investment (ROI) measures the [gain](#) or loss generated on an investment relative to the amount of money invested. ROI is usually expressed as a percentage and is typically used for personal financial decisions, to compare a company's profitability or to compare the efficacy of different investments.

Formula:

$$\text{ROI} = (\text{Net Profit} / \text{Cost of Investment}) \times 100$$

Example:

The ROI calculation is flexible and can be manipulated for different uses. A firm may use the calculation to compare the ROI on different potential investments.

For example, an investor invests \$1,000 and gets a return of \$1,200 in one year. The [net profit](#) from the investment would be \$200 and the ROI would be calculated as follows:

$$\text{ROI} = (200 / 1,000) \times 100 = 20\%$$

The ROI in the example above would be 20%. The calculation can be altered by deducting [taxes](#) and fees to get a more accurate picture of the total ROI.

Return on Investment (ROI) is the benefit to an investor.

Profitability index (PI)

Profitability index (PI), also known as profit investment ratio (PIR) and value investment ratio (VIR). It is the ratio of payoff to investment of a proposed project. It is a useful tool for ranking projects because it allows you to quantify the amount of value created per unit of investment.

The ratio is calculated as follows:

- Assuming that the [cash flow](#) calculated does not include the investment made in the project, a profitability index of 1 indicates breakeven. Any value lower than one would indicate that the project's present value ([PV](#)) is less than the initial investment. As the value of the profitability index increases, so does the financial attractiveness of the proposed project.

Rules for selection or rejection of a project:

- If $PI > 1$ then accept the project
- If $PI < 1$ then reject the project

For example:

Investment = \$40,000

Life of the Machine = 5 Years

Year	Cash in - flow
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1	18000
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2	12000
---	-------

3	10000
---	-------

4	9000
---	------

5	6000
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Calculate [Net present value](#) at 10% and PI:

Year	Cash in-inflow	PV@10%	PV
1	18000	0.909	16362
2	12000	0.827	9924
3	10000	0.752	7520
4	9000	0.683	6147
5	6000	0.621	3726

Total Present Value	\$ 43,679
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(-) Investment	\$ 40,000
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NPV	\$ 3,679
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$PI = 43679/40000 = 1.091 > 1 \Rightarrow$ Accept the project